STANDARD OPERATING PROCEDURE OF Electric Shock Resistance VF-J01

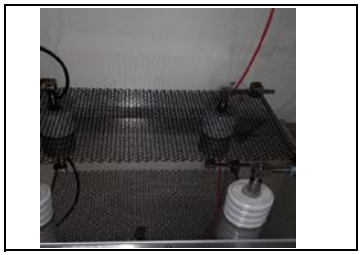
Lab /SOP/ 50



1.ESR Machine that meets CSA Z195 and ASTM F - 2412:2011 requirements.
2.Solid metal (electrically conductive) spheres of 3 mm in diameter
3.Aluminum plate of 100 mm diameter @ 1 kg mass
4.Stopwatch or timer
5.Steel Ruler



Place the inner boot electrode that consist of solid metal (electrically conductive) spheres of 3 mm in diameter (e.g., BB shot) inside the footwear to be tested. It should cover the entire inner sole surface of the footwear to a depth of not less than 30 mm.



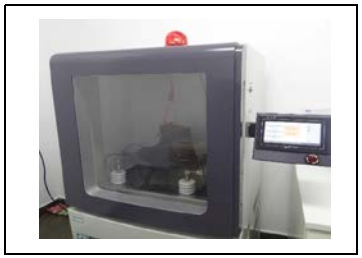
Check the metal mesh, should be mild steel flexible fire screen, size 400 x 220 mm.
Check the surface of the mesh screen. It should be flat surface and parallel with the testing base.



Ensure that the moderate tension is achieved by positioning at the center the 1 kg mass aluminum plate of 100 mm diameter. The standard deflection should be 15 +/- 2.5 mm.



The inner electrode shall be maintained at ground potential. The test voltage shall be applied to the outer electrode at a low level (approximately 0 V) and steadily increased at a maximum rate of 1 kV/s until the prescribed test voltage is reached. This voltage shall be maintained for 1 minute.



The current shall be recorded at the beginning and at the end of the test period.



Leakage current /mA

Prepared by	Approved by	REVISION		DATE	